**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Long consecutive runs of identical symbols.

Q2. 24.

Q3. RAM.

Q4. Sum 0, carry-out 1.

### You Do pages · Independent questions

Q1. Repeated substrings replaced by shorter references [3]; mapping resolves tokens for exact restoration [3].

Q2. 1 2 1 1 [3]; BLUE RED [3], with token separators assumed.

Q3. Repeated sequences help [2], tokens shorten recurring data [2], dictionary overhead matters [2], little redundancy in high-entropy data and justified choice [2].

### Exit Ticket · Model answer

False. The dictionary allows exact replacement when decoding.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 129 · Lesson 129 · Dictionary Compression and Repeated Substrings

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. RLE is effective on what kind of input?

Q2. How many bits are in three bytes?

Q3. Which memory is typically volatile: RAM or ROM?

Q4. What is the binary result of 1 + 1 in a full adder with carry-in 0?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

dictionary coding — Encodes recurring sequences by referring to dictionary entries.

token — Short code representing a longer sequence.

overlap — Repeated sequence used by more than one part of the source.

lossless — Reconstruction is exact when dictionary and token mapping are known.

### New learning

Find recurring text — Locate useful repeated words or substrings.

Build dictionary — Assign tokens to recurring sequences and store mapping.

Encode and decode — Replace sequences with tokens; decode using the same mapping.

### Worked example

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

1. Two dictionary entries — Predict the first step from the prompt.

2. #1 #2 #1 — Explain the intermediate result or decision.

3. Restore original phrase — Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

## We Do · Practise with guidance

Dictionary {A:'ABCD'}; encode 'ABCDABCDXYZABCD' and discuss the result.

ABCD repeats three times

Predict the first step from the prompt.

Token A for ABCD

Explain the intermediate result or decision.

Count dictionary overhead

AAXYZA, using single-character token A and leaving XYZ as literals under this toy scheme. Original 15 characters become 6 token/literal characters, plus dictionary overhead; real formats require unambiguous token conventions.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain dictionary compression and why the decoder needs the dictionary. [6]

### Question 2

Using entries {1:'RED',2:'BLUE'}, encode 'RED BLUE RED RED' and decode '2 1'. [6]

### Question 3

Evaluate dictionary coding for a document with frequent terms versus already-compressed random-looking data. [8]

## Exit Ticket · Answer before checking

Dictionary coding requires permanently discarding repeated words.